

ECE 447

Lesson 19

Pulse Code Modulation



UNITED STATES
AIR FORCE
ACADEMY

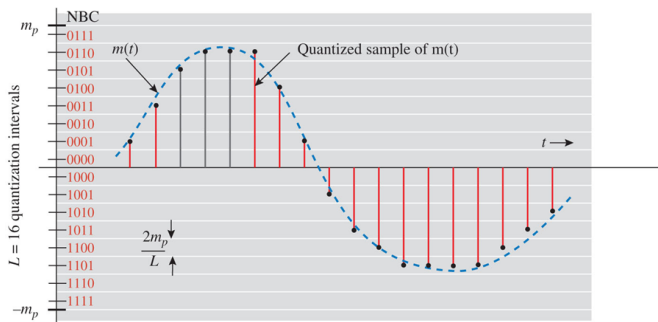
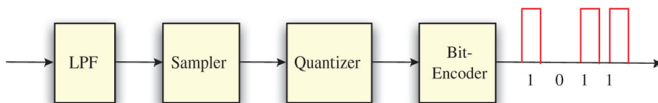
SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **Lab 4.** Due tonight (Lesson 19, 30 Sep) to Gradescope.
 - **HW4.** Due Lesson 21 (Tue, 6 Oct) to Gradescope.
 - **Lab 5.** Begins Lesson 21 (Tue, 6 Oct); due Lesson 24 (Thu, 15 Oct).

EXAMPLE PROBLEM - 5.1-7

SAMPLING COMPUTER EXAMPLE

PULSE CODE MODULATION



QUANTIZATION ERROR AND SNR

$$\begin{aligned}\left(\frac{S_o}{N_o}\right)_{\text{dB}} &= 10 \log_{10} \left(\frac{S_o}{N_o}\right) \\ &= 10 \log_{10} \left[c(2)^{2n}\right] \\ &= 10 \log_{10} c + 2n \log_{10} 2 \\ &= (\alpha + 6n) \text{ dB}\end{aligned}$$

PCM COMPUTER EXAMPLE